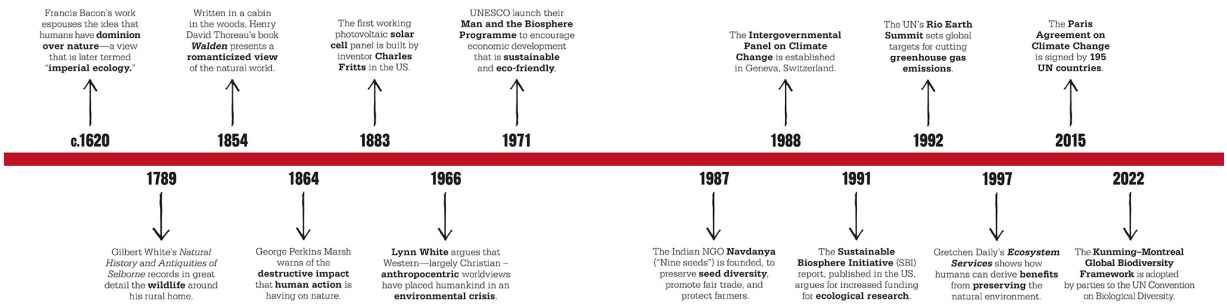
INTRODUCTION

Early in the 17th century, English philosopher and scientist Francis Bacon wrote of the need to control and manage nature. By the end of the 18th century, in contrast, English vicar Gilbert White was writing in favor of a peaceful coexistence between people and the natural world. Yet in his lifetime, powerful new steam engines unleashed the ravages of industrialization—the reaction against which would later provide a major impetus for the environmental movement.

Possibly the first systematic analysis of humanity's destructive impact was American diplomat George Perkins Marsh's 1864 book, *Man and Nature*. Marsh warned, among other things, that deforestation could lead to the creation of deserts, and he pointed out that resource scarcity was generally the result of human actions rather than natural causes.



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Renewable and clean

Before the Industrial Revolution, most energy had been renewable—the energy of human and animal labor, wind- and watermills, and sustainable wood. From the mid-18th century there was a dramatic shift to coal. The most efficient fuel for firing furnaces and factories, it came at a price—choking pollution and the then-unknown rise in atmospheric greenhouse gases.